Lab 6 Report Microprocessor

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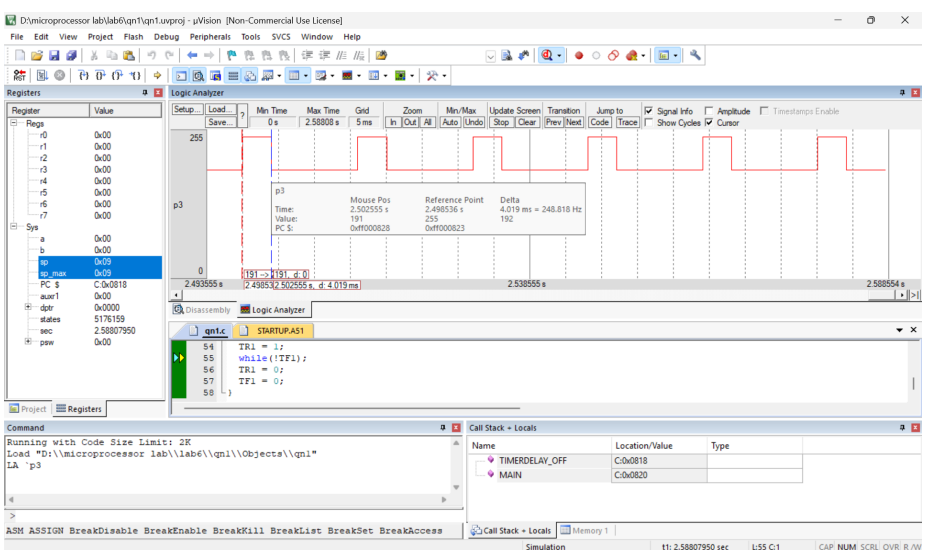
1 Design and Implementation of an Unsymmetrical Square Waveform Generator

1.1 Objective

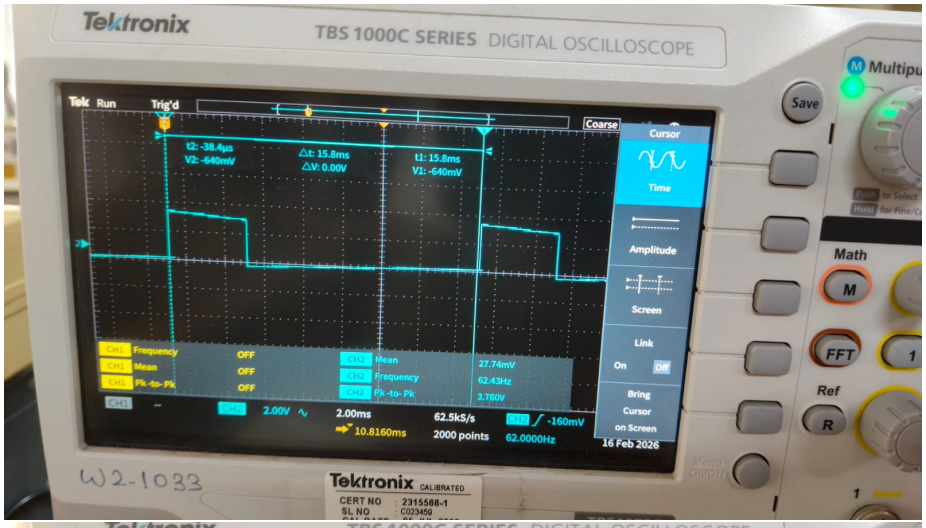
To generate an unsymmetrical square wave with a specific duty cycle (4 ms ON, 12 ms OFF) at port pin P3.6 using the 8051 Timer 1 in polling mode, and to verify the waveform timing using both Keil Logic Analyzer and a physical DSO.

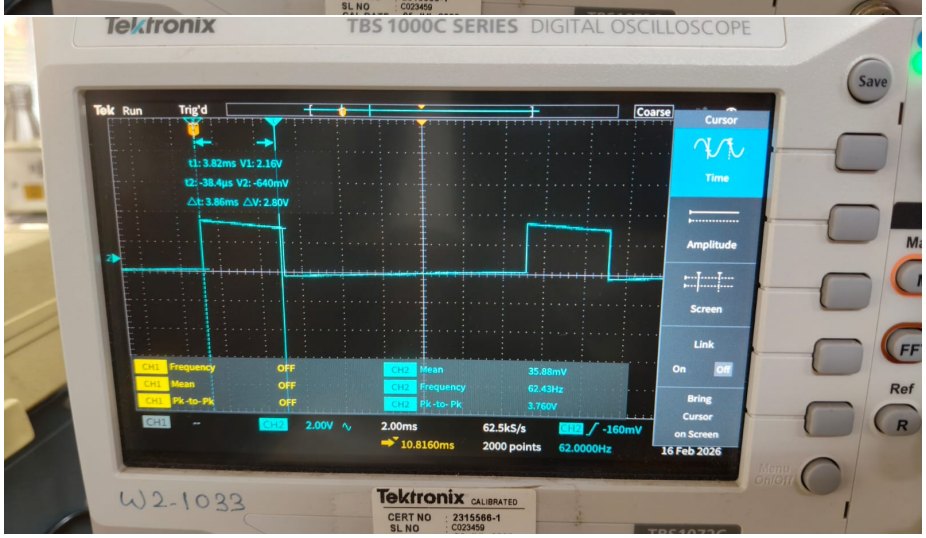
1.2 Observation



- Keil Logic Analyzer: The simulated waveform shows a logic high duration of 4.0 ms and a logic low duration of 12.0 ms. The total period is observed to be 16.0 ms, corresponding to a frequency of 62.5 Hz
- DSO (Pt-51 Board): Upon flashing the code, the DSO displays a rectangular wave at P3.6. Measurement cursors confirm the timings align with

the calculated values, with slight overhead due to instruction execution time in the polling loop





1.3 Conclusion

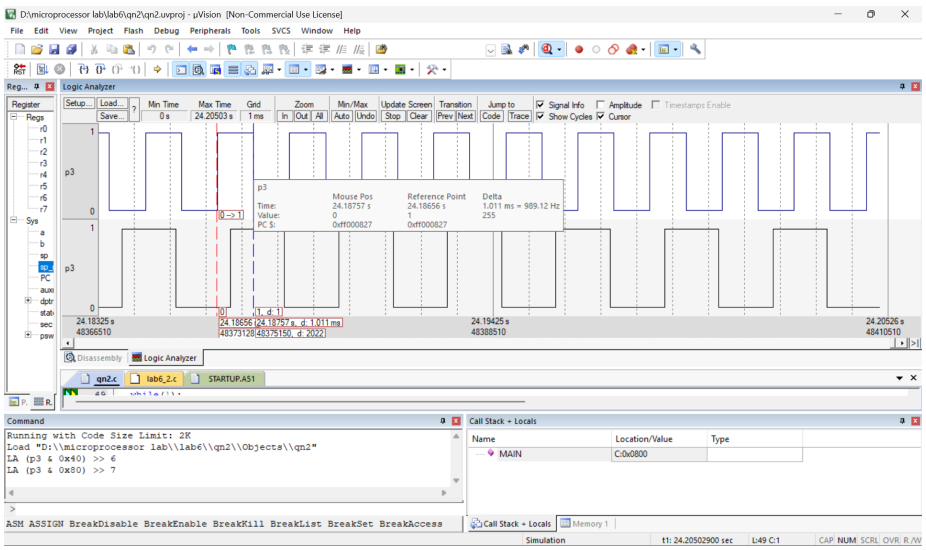
The 8051 timers provide an accurate method for generating non-50% duty cycle signals without the overhead of interrupts. By calculating the 16-bit reload values based on the machine cycle ($1.085\mu s$), precise pulse-width modulation (PWM) can be achieved for embedded applications.

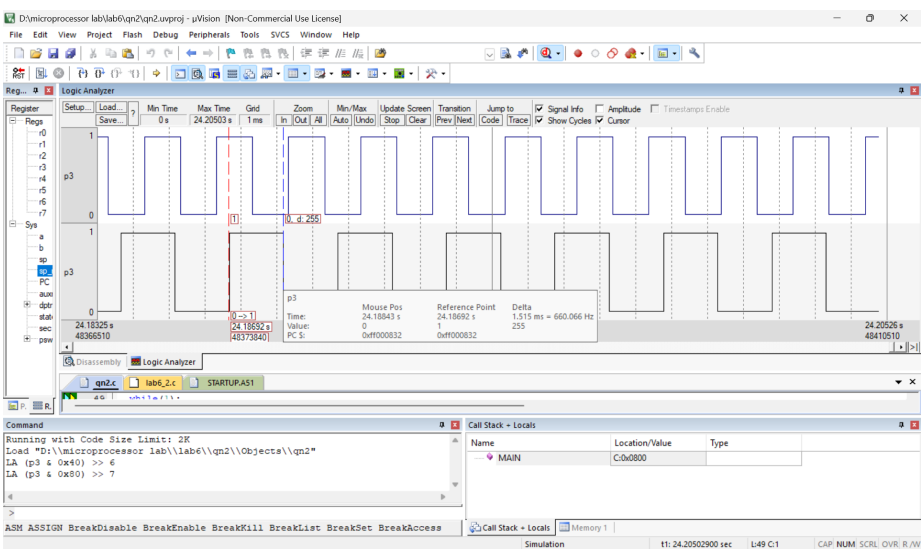
2 Simultaneous Generation of Multiple Square Waves using 8051 Timers in Interrupt Mode

2.1 Objective

To implement and verify the simultaneous generation of two independent square waves (2 ms and 3 ms periods) at pins P3.6 and P3.7 using Timer 0 and Timer 1 in Interrupt Mode on an 8051 microcontroller.

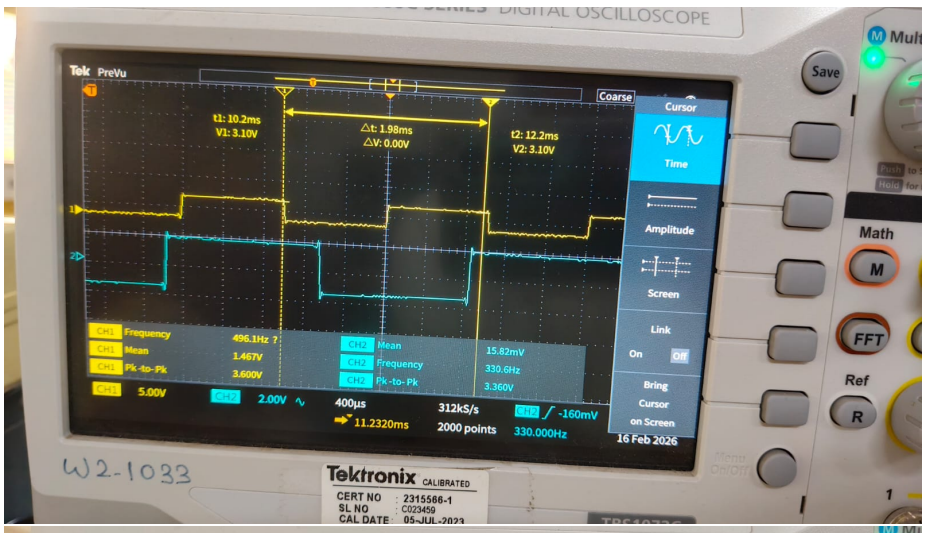
2.2 Results

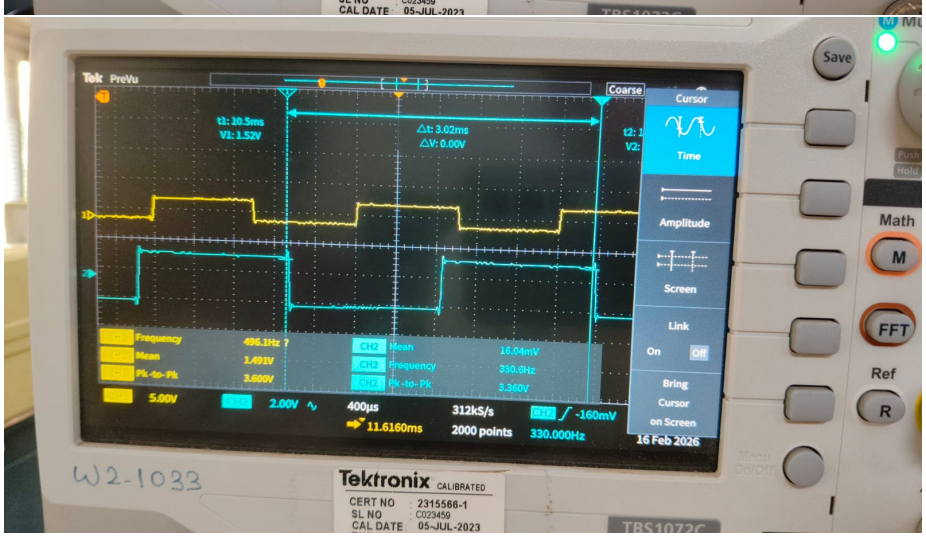




2.3 Observations

- Two distinct square waves are visible. Signal 1 (P3.6) shows a transition every 1 ms (total period 2 ms), and Signal 2 (P3.7) shows a transition every 1.5 ms (total period 3 ms).
- Using a dual-trace setup on the DSO, both waveforms appear stable. The interrupts handle the toggling seamlessly, with the 2 ms wave showing a frequency of 500 Hz and the 3 ms wave showing 333.3 Hz.





2.4 Conclusion

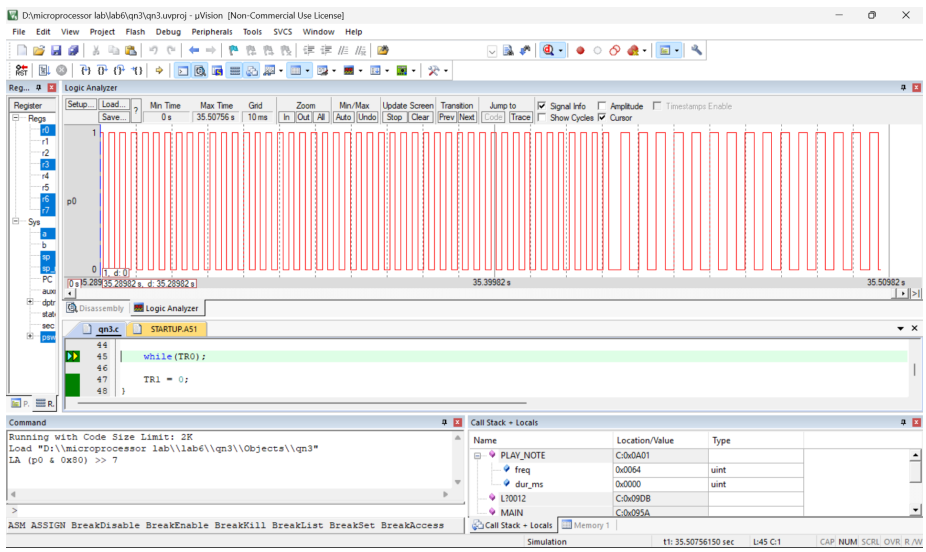
Interrupt-driven timing is superior to polling for multi-tasking. By using Timer ISRs, the 8051 can manage multiple time-critical outputs simultaneously without one process blocking the other. The 16-bit Mode 1 provides the necessary resolution for millisecond-level precision.

3 Implementation of a Musical Note Generator using 8051 Dual-Timer Interfacing and Speaker Interfacing

3.1 Objective

To generate a sequence of musical notes by controlling square wave frequencies with Timer 0 and note durations with Timer 1, outputting the signal to P0.7 for speaker playback.

3.2 Observations



- P0.7 shows varying pulse widths corresponding to frequencies like 480 Hz and 226 Hz, with the signal persisting for exact intervals such as 300 ms or 600 ms.
- When connected to the driver circuit, the speaker produces distinct audible tones. The melody played is the theme from "Stranger Things" (or a very similar eerie synth-style progression common in your recent interests).

3.3 Conclusion

By using nested timer interrupts, the 8051 can effectively manage two time-bases: a high-frequency time-base for pitch and a low-frequency time-base for rhythm. The SL100 transistor acts as a necessary current amplifier to drive the low-impedance speaker from the high-impedance MCU port.